

# Test 1 - Autonomous Incident Investigation Report

## 1. Executive Summary

The supplied evidence shows a sequence of activity on **DESKTOP-5OG7C28**, monitored by the Wazuh agent **Windows\_VM (192.168.232.130)**, involving PowerShell and the Windows Command Shell under the account **DESKTOP-5OG7C28\Darragh Casey**.

The principal finding is that an elevated PowerShell process launched **cmd.exe**, which was instructed to execute the batch file:

```
C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat
```

At **2026-10-02 14:55:27.411 UTC**, Sysmon recorded **cmd.exe** PID **8188** with PowerShell PID **1212** as its direct parent. Both processes ran in the same user context, and the command shell operated at **High integrity**. The parent PowerShell command explicitly used **Start-Process** against the same batch file. This parent-child relationship is directly supported by matching process identifiers and command-line telemetry.

Several **hostname.exe** and **whoami.exe** executions occurred immediately before this execution, together with PowerShell checks for the existence of the same batch file. These events are temporally and contextually associated with the activity, although the supplied evidence does not establish a direct parent-child relationship between all of them.

The evidence supports **command/script execution rather than a multi-stage intrusion**. There is no supplied evidence demonstrating persistence, credential theft, lateral movement, command-and-control communication, payload download, or other subsequent compromise.

The Wazuh alerts categorising temporary PowerShell-created **.ps1** files as **T1105 - Ingress Tool Transfer** are not independently supported by the underlying telemetry. The records prove local file creation but do not demonstrate that those files were transferred onto the system.

## 2. Scope of Activity

The evidence supports a **single primary execution sequence**, accompanied by likely preparatory or environmental checks.

- PowerShell checking whether **T1059.003\_script.bat** exists.
- PowerShell subsequently using **Start-Process** against that batch file.
- PowerShell spawning **cmd.exe**.
- **cmd.exe** being invoked with **/c** to execute **T1059.003\_script.bat**.

The evidence also contains executions of **hostname.exe** and **whoami.exe** shortly before the primary execution. These establish that system and user information was queried, but their precise relationship to the subsequent PowerShell to **cmd.exe** chain cannot be proven because their Sysmon records contain a zeroed **ParentProcessGuid** and do not identify the responsible parent image.

The supplied telemetry does **not** show what commands were contained inside **T1059.003\_script.bat**. No subsequent child processes attributable to PID 8188 are present in the supplied evidence. Therefore, the investigation cannot determine whether the batch file performed additional actions that were not captured within this evidence window.

Background Windows activity, including Microsoft Edge, OneDrive, BITS and Windows Update processes, was also observed. No evidence establishes a relationship between those events and the command-shell execution.

### 3. Detailed Investigation

#### Initial system and user checks

At **14:52:59.793 UTC**, Sysmon Event ID 1, record **15234**, recorded:

`C:\Windows\System32\HOSTNAME.EXE`

The process ran as **DESKTOP-5OG7C28\Darragh Casey**, at High integrity, from:

`C:\Users\Darragh Casey\Desktop\T1059.003\`

At **14:52:59.820 UTC**, record **15235** recorded:

`C:\Windows\System32\whoami.exe`

It ran under the same account, logon ID **0xA8D6D**, terminal session 1 and High integrity level.

These executions directly establish that the hostname and current-user information were queried. Their purpose is not explicitly recorded.

**Inference:** Their proximity, identical user/session context and working directory make them plausibly associated with the subsequent activity. However, their direct relationship to the later command-shell execution cannot be proven from the available parent-process fields.

#### PowerShell file-existence check

At **14:53:00.058 UTC**, Sysmon record **15236** recorded PowerShell PID **7164** executing a command equivalent to:

```
if (Test-Path "C:\AtomicRedTeam\atomics\..\ExternalPayloads\T1059.003_script.bat") {exit 0}
else {exit 1}
```

This directly shows PowerShell testing whether the batch file existed.

At **14:53:00.175 UTC**, the same PowerShell process created:

`C:\Users\Darragh Casey\AppData\Local\Temp\__PSScriptPolicyTest_jarjq0wa.5fg.ps1`

Wazuh generated a level-15 alert describing this as an executable being dropped in a malware-associated directory and mapped it to T1105.

The underlying record, however, only establishes that **powershell.exe** created a **.ps1** file named **\_\_PSScriptPolicyTest...** in the user's temporary directory. It contains no evidence of a download, remote source, transfer mechanism or network connection associated with that file.

Therefore, **T1105 cannot be established from this event alone**.

#### Activity immediately preceding execution

A second cluster occurred approximately two and a half minutes later.

Between **14:55:25.945** and **14:55:26.060 UTC**, three instances of **HOSTNAME.EXE** were recorded.

At **14:55:26.091 UTC**, **whoami.exe** was executed.

These processes again ran as **DESKTOP-5OG7C28\Darragh Casey**, with High integrity and logon ID **0xA8D6D**.

At **14:55:26.328 UTC**, PowerShell PID **4484** again tested for:

`C:\AtomicRedTeam\atomics\..\ExternalPayloads\T1059.003_script.bat`

At **14:55:26.443 UTC**, that PowerShell process created another temporary file:

`__PSScriptPolicyTest_z4baxwgm.atc.ps1`

Wazuh again raised the T1105-oriented level-15 alert. As with the earlier occurrence, the underlying record establishes file creation but does not demonstrate ingress tool transfer.

Another **HOSTNAME.EXE** execution occurred at **14:55:26.679 UTC**, followed by **whoami.exe** at **14:55:26.710 UTC**.

## PowerShell launches the batch file

At **14:55:26.958 UTC**, Sysmon record **15248** recorded PowerShell PID **1212** executing:

```
Start-Process "C:\AtomicRedTeam\atomics\...\ExternalPayloads\T1059.003_script.bat"
```

The process ran as:

```
DESKTOP-50G7C28\Darragh Casey
```

- Integrity level: High
- Logon ID: 0xA8D6D
- Terminal session: 1

At **14:55:27.119 UTC**, the same PowerShell process created:

```
C:\Users\Darragh Casey\AppData\Local\Temp\__PSScriptPolicyTest_sct5nwuc.3fq.ps1
```

A third Wazuh T1105-oriented alert was generated.

## Windows Command Shell execution

At **14:55:27.411 UTC**, approximately 453 milliseconds after the PowerShell process was created, Sysmon record **15251** recorded:

```
C:\Windows\System32\cmd.exe
```

PID:

```
8188
```

Command line:

```
C:\WINDOWS\system32\cmd.exe /c ""C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat" "
```

Its parent was explicitly recorded as:

```
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
```

Parent PID:

```
1212
```

Parent Process GUID:

```
{4b2ce92d-c5de-6abf-7e04-000000001400}
```

This exactly matches the PowerShell process recorded immediately beforehand.

The parent command line was also recorded as the **Start-Process** command targeting the same batch file.

Consequently, the relationship **PowerShell PID 1212 -> cmd.exe PID 8188 -> execution request for T1059.003\_script.bat** is a **directly observed fact**, not an inference. Wazuh generated rule **92004**, "Powershell process spawned Windows command shell instance," for this event.

The command shell ran with **High integrity**, meaning the observed execution was occurring in an elevated context.

The supplied evidence ends without showing the contents of the batch file or any subsequent process generated by PID 8188. It is therefore not possible to determine from this dataset what further operations, if any, the batch file performed.

## 4. Evidence Analysis

The strongest evidence is Sysmon **Event ID 1**, process-creation telemetry.

The decisive record is **15251**, because it provides the complete execution relationship:

- Image: C:\Windows\System32\cmd.exe

- PID: 8188
- Command: cmd.exe /c ...T1059.003\_script.bat
- User: DESKTOP-5OG7C28\Darragh Casey
- Integrity: High
- Parent image: powershell.exe
- Parent PID: 1212
- Parent Process GUID: {4b2ce92d-c5de-6abf-7e04-000000001400}
- Parent command: PowerShell Start-Process targeting the same batch file.

Record **15248** independently identifies PID 1212 and the corresponding PowerShell **Start-Process** command, providing corroboration for the relationship.

The **HOSTNAME.EXE** and **whoami.exe** records establish additional command execution but contain insufficient parent information to prove that they belong to the same process tree. Their temporal proximity, user context, logon session and working directory provide circumstantial support for a relationship.

## Assessment of Wazuh alerts

The Wazuh Windows Command Shell alert is well supported by its underlying Sysmon record. The alert correctly identifies PowerShell spawning **cmd.exe**, and the actual command line proves command-shell use.

By contrast, the three Wazuh level-15 alerts categorising the **\_\_PSScriptPolicyTest\_\*.ps1** creations as **T1105 / Ingress Tool Transfer** should not be treated as proof of that technique. The underlying Event ID 11 records only demonstrate file creation by PowerShell.

No supporting evidence of an external source, download utility, network transfer or remote connection involving these files is present.

## Excluded or unrelated evidence

The dataset also contains:

- Microsoft Edge Update execution.
- OneDrive updater execution and network activity.
- DNS queries associated with OneDrive/system services.
- BITS changing from demand start to automatic and subsequently back to demand start.
- Windows security logon/group-enumeration events involving SYSTEM.
- Windows taskhostw.exe and SoftLandingTask.exe.
- smartscreen.exe.
- MoUsoCoreWorker.exe.

These events do not share the process chain, Process GUIDs or command lines associated with the PowerShell/command-shell activity.

The BITS state changes occurred around OneDrive updater activity. The available evidence does not prove causation, but this contextual relationship makes them more consistent with unrelated background activity than with the identified command-shell execution. The service-change alerts themselves contain no direct link to PID 1212 or PID 8188.

## 5. Timeline of Events

**14:52:59.793 UTC** - HOSTNAME.EXE executes as Darragh Casey, High integrity.

**14:52:59.820 UTC** - whoami.exe executes under the same account/session.

**14:53:00.058 UTC** - PowerShell PID 7164 checks whether T1059.003\_script.bat exists.

**14:53:00.175 UTC** - PID 7164 creates \_\_PSScriptPolicyTest\_jarjq0wa.5fg.ps1.

**14:55:25.945-14:55:26.060 UTC** - Three additional HOSTNAME.EXE processes execute.

**14:55:26.091 UTC** - whoami.exe executes.

**14:55:26.328 UTC** - PowerShell PID 4484 again checks whether T1059.003\_script.bat exists.

**14:55:26.443 UTC** - PID 4484 creates \_\_PSScriptPolicyTest\_z4baxwgm.atc.ps1.

**14:55:26.679 UTC** - Another HOSTNAME.EXE process executes.

**14:55:26.710 UTC** - Another whoami.exe process executes.

**14:55:26.958 UTC** - PowerShell PID 1212 executes Start-Process targeting T1059.003\_script.bat.

**14:55:27.119 UTC** - PID 1212 creates \_\_PSScriptPolicyTest\_sct5nwuc.3fq.ps1.

**14:55:27.411 UTC** - cmd.exe PID 8188 is created by PowerShell PID 1212 and executes the batch file using /c.

The Sysmon **UtcTime** values above are UTC. Wazuh's outer event timestamps are recorded with the **+0200** offset; for example, the command-shell event appears as approximately **16:55:27 local (+02:00)** while its Sysmon execution time is **14:55:27 UTC**.

## 6. MITRE ATT&CK Mapping

### T1059.003 - Command and Scripting Interpreter: Windows Command Shell

**Confidence: High**

Directly supported.

**cmd.exe** was invoked with **/c** to execute a **.bat** file:

```
cmd.exe /c "C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat" "
```

The process and its PowerShell parent are directly recorded by Sysmon. Wazuh independently triggered rule 92004 for the same underlying event.

### T1059.001 - Command and Scripting Interpreter: PowerShell

**Confidence: High**

PowerShell is directly observed executing multiple commands, including **Test-Path** checks and a **Start-Process** operation that results in the creation of **cmd.exe**.

This establishes actual use of PowerShell as a command execution mechanism.

### T1082 - System Information Discovery

**Confidence: Medium**

Multiple executions of **HOSTNAME.EXE** are directly observed. Querying the host name constitutes collection of system-identifying information.

The executions themselves are facts; their inclusion in the same investigative sequence is assessed with Medium rather than High confidence because the parent process is not available in those records.

### T1033 - System Owner/User Discovery

**Confidence: Medium**

Multiple executions of **whoami.exe** are directly observed under the investigated account.

Again, execution of the command is directly observed, while its exact relationship to the PowerShell to command-shell chain is inferred from timing and context.

## T1105 - Ingress Tool Transfer

### Not established by the supplied evidence

Wazuh mapped the temporary \_\_PSScriptPolicyTest\_\*.ps1 creation events to T1105. However, the underlying records show file creation only.

There is no evidence demonstrating:

- an external source,
- a remote host,
- a download,
- a network transfer associated with the files, or
- another mechanism by which a tool entered the system.

Therefore, the investigation does **not** adopt the alert's T1105 classification as a supported finding.

## 7. Overall Investigation Conclusion

The supplied evidence establishes with **High confidence** that an elevated PowerShell process caused the Windows Command Shell to execute:

```
C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat
```

on **DESKTOP-5OG7C28**, under **DESKTOP-5OG7C28\Darragh Casey**, at approximately **14:55:27 UTC on 2 October 2026**.

The PowerShell-to-cmd.exe relationship is directly supported through matching process identifiers, Process GUIDs, parent-process data and command lines.

Additional **hostname.exe**, **whoami.exe** and PowerShell file-existence checks occurred immediately before the principal execution. These are plausibly associated with the same activity, although their complete process ancestry is not available.

The evidence supports **command execution and associated discovery activity**, but does not support classification of the supplied activity as a broader multi-stage intrusion. There is no evidence in the provided records of persistence, credential access, lateral movement, exfiltration, command-and-control activity or payload transfer.

The contents of **T1059.003\_script.bat** are not supplied, and no downstream process activity from **cmd.exe** is visible. Consequently, the actual internal operations or effect of the batch file cannot be determined from this evidence.

Because the activity occurred in an **isolated laboratory environment**, the technical behaviours should not by themselves be interpreted as proof of a genuine hostile compromise. The evidence establishes what occurred technically, but not malicious intent.

**Overall investigation confidence: High** for the principal PowerShell to Windows Command Shell execution; **Medium** for associating the surrounding hostname.exe and whoami.exe activity with that execution; and **Low/insufficient evidence** for claims about what the batch file subsequently performed or for the Wazuh T1105 classification.